SNEC · SINGAPORE NATIONAL EYE CENTRE

TRAINING & EDUCATION

# EyeBot

*An AI training companion for the whole of ophthalmic education — grounding every learner, from first-year student to practising clinician, in faithful, adaptive, measurable practice.*

★ PREPARED FOR THE 2026 INTERNATIONAL E-LEARNING AWARDS · ACADEMIC DIVISION · E-LEARNING EXPERIENCES

PREPARED FOR
Dr JB · Director, SNEC Training & Education

PREPARED BY
Caleb Teo · SNEC Training & Education

DOCUMENT
June 2026 · Confidential Draft v3.0

## 01 EXECUTIVE SUMMARY

Imagine every ophthalmic trainee — whatever their role — with a tutor who never sleeps, a patient who is always available, and a curriculum that adapts to exactly what they keep forgetting.

**SNEC EyeBot** is an AI learning companion that transforms a clinical curriculum into three evidence-based experiences — a Socratic tutor, inexhaustible virtual-patient simulations, and a self-optimising memory engine — woven into one adaptive loop. Grounded today in SNEC's clinical curriculum and architected to serve the full spectrum of eye-care learners, it also gives educators a live, cohort-wide view of mastery for the first time. The platform is built, demonstrable, and ready to pilot.

### 3-in-1

Tutor, clinical simulator and memory engine working as a single adaptive loop.

### Grounded

Every answer drawn from validated clinical material — faithful, not the open internet.

### Cohort

Live at-risk detection, retention scoring and weekly digests for educators.

A fixed curriculum becomes a living, personalised, measurable learning experience — and positions SNEC at the front of AI-enabled clinical education.

02 THE CHALLENGE

# Training the eyes behind the eye care.

Across the eye-care continuum — medical students on rotation, optometry students, ophthalmology residents, nurses and allied-health technicians — training is rigorous, yet it meets four structural limits that no amount of dedication can fully solve.

### Expert time is scarce

One educator cannot give every trainee unlimited one-to-one tutoring or on-demand case practice.

### Case exposure is uneven

Which conditions a trainee actually meets is left to the luck of the clinical roster.

### Knowledge decays

Without deliberate reinforcement, much of what is taught is forgotten within weeks of the lesson.

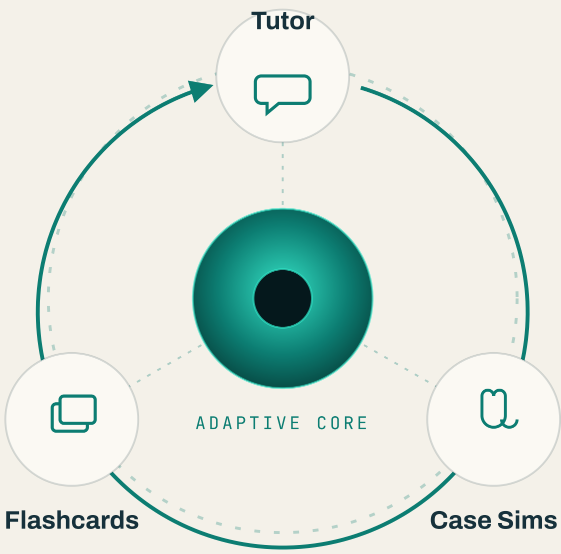
### Cohort gaps are invisible

Educators often discover weaknesses at assessment time — when it is already late to intervene.

## EyeBot is engineered to dissolve all four — at once.

03 THE SOLUTION

# One companion. Three modes. A single adaptive loop.




MODE 01

### Socratic Tutor

Plain-language questions met with Explanation → Mechanism → Clinical Pearl → a check for understanding, always grounded in the clinical knowledge base.

MODE 02

### Case Simulations

Interview a virtual patient, order investigations, commit to a diagnosis and plan — graded on a four-domain rubric out of 40 with instant feedback.

MODE 03

### Spaced Flashcards

High-yield cards auto-generated from each session and every missed clinical point, scheduled by the SM-2 algorithm for durable recall.

...and quietly, an analytics layer watches the whole cohort — turning individual practice into institutional insight.

04 THE INNOVATION

# An innovation — not an integration.

EyeBot is not a chatbot bolted onto a syllabus. Five deliberate advances make it genuinely new for clinical education.

## 01 Curriculum-grounded clinical AI

Answers are retrieved from the institution's own validated material — faithful and auditable, never the untraceable guesswork of a generic model.

## 02 Inexhaustible virtual patients

The AI plays both patient and examiner: unlimited, standardised case exposure with rubric-graded feedback — independent of roster luck.

## 03 A self-optimising loop

Each learner's weak points automatically become their next flashcards and tougher cases — genuine personalisation, at scale.

## 04 Reliable by architecture

A Workflows–Agents–Tools design separates AI judgement from deterministic execution — trustworthy enough for clinical training.

## 05 Cohort intelligence

Learning analytics purpose-built for clinical education: at-risk detection, retention scoring and learning-velocity — live, not at term's end.

**Together: the first AI companion built for how eye care is actually taught.**

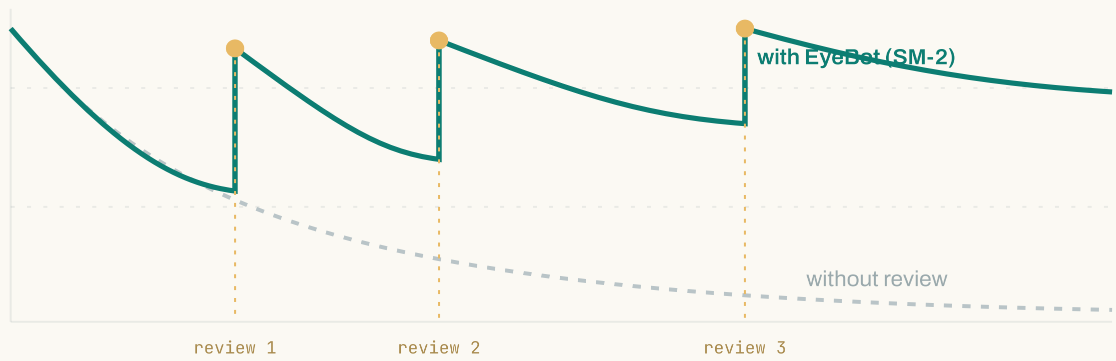
05 PEDAGOGICAL FOUNDATION

# Grounded in learning science, not novelty.

Every design decision in EyeBot maps to an established principle of how clinicians actually learn and retain.

FIG. 1 — SPACED REPETITION VS. THE FORGETTING CURVE

Retention ↑ · Time →




### Retrieval & spacing

SM-2 spaced repetition fights the forgetting curve — each review lengthens durable retention.

### Socratic scaffolding

The tutor explains mechanism, then checks understanding — active recall over passive reading.

### Case-based reasoning

Simulations rehearse the full clinical chain: history, investigation, diagnosis, management.

### Formative feedback

Rubric scores and targeted cards turn every mistake into the learner's next rep.

06 BUILT FOR EVERYONE IN EYE CARE

# One engine. Every learner.

EyeBot adapts its content, cases and rubric to each role — and the same engine extends across specialties and institutions. What begins with SNEC's programme scales to the whole of ophthalmic education.

**Medical Students**
Eye-rotation foundations

**Optometry Students**
Refraction & primary care

**Ophthalmology Residents**
Clinical decision-making

**Ophthalmic Nurses**
Peri-operative & patient care

**Allied Health · OA / OT / PSA**
Technical & assisting skills

### Role-aware

Content, case libraries and rubrics tuned to each learner's scope of practice.

### Specialty-extensible

The same engine accepts new curricula — glaucoma, retina, paediatrics and beyond.

### Institution-ready

PDPA-compliant consent, bcrypt credentials and role-based access — privacy by design.

From a single programme to an institution-wide standard for how eye-care professionals are trained.

07 PILOT PLAN & AWARD FIT

# From prototype to proof.

## PROPOSED SNEC PILOT

**COHORT**

A representative intake of ophthalmic trainees over an 8–12 week module, benchmarked against a prior cohort as historical control.

**METRICS**

Pre/post knowledge test · case-rubric progression · flashcard retention · engagement (active days & streaks) · educator time saved · learner satisfaction.

**METHOD**

Baseline → guided daily use → measured uplift, with the analytics layer capturing evidence automatically throughout.

## FIT TO IELA CRITERIA

| ATTRIBUTE | HOW EYEBOT DELIVERS |
|---|---|
| **Educational soundness** PEDAGOGY | Curriculum-grounded; SM-2 spacing; Socratic + case-based design rooted in learning science. |
| **Effectiveness** OUTCOMES | Closes gaps in a measured loop; analytics quantify uplift at learner and cohort level. |
| **User experience** ENGAGEMENT | Polished, gamified and mobile-friendly, with daily check-ins that build habit. |
| **Significance** IMPACT | First AI companion spanning the ophthalmic learning continuum; scalable across SingHealth. |

Because impact is *measured, not asserted*, the pilot doubles as the evidence base for the award submission itself.

08 ROADMAP & VISION

# Where this goes next.

Q3 2026
**SNEC Pilot**
Run with a live intake; capture the outcome evidence.

Q4 2026
**Multi-role rollout**
Extend across every eye-care learner type.

2027
**Specialty expansion**
New curricula, richer simulation media, UI refresh.

BEYOND
**SingHealth-wide**
Other institutions on the same engine.

SNEC EyeBot turns a fixed curriculum into a living, personalised and measurable learning experience — and positions SNEC at the forefront of AI-enabled clinical education.

ENGINEERED WITH

FastAPI · Supabase · React + TypeScript · Anthropic Claude · Workflows · Agents · Tools